

COSME DANIEL AQUINO OVELAR

◇ Bonn, Germany

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EDUCATION

Master of Science in Astrophysics, Rheinische Friedrich-Wilhelms-Universität (Universität Bonn). 2024-2026

Bachelor of Science in Astrophysics, University of Oklahoma.

May 2024

Minor in Mathematics and Certificate in Data Analytics.

Noteworthy Classes Taken: Advanced Applied Statistics, Applied Statistical Methods, and Linear Algebra.

SKILLS

Programming Languages	Python, R and Mathematica
Languages	Spanish, English

RESEARCH PROJECTS

Optical and Gamma-ray Variability of AGN and Gamma-ray Binaries

2022-2024

Supervisor: Dr. Xinyu Dai.

- Implemented TESS pipeline to extract light curves of more than fifteen bright blazars from the optical survey to look for stochastic variability.
- Utilized Python packages like Numpy, Scipy, Astropy, and R for data cleaning.
- Performed cross-matching techniques using SQL to find potential AGN targets in 2MASS and SDSS surveys.
- Implemented Lomb-Scargle Analysis to Fermi-LAT and ASAS-SN data.
- Familiarized with cross-correlation analysis techniques, typically performed in AGN, to apply it to a different type of objects, and study time-lag correlations using [JAVELIN](#) software.
- Applied statistical models to model gamma-ray flares and measure photon flux from Fermi-LAT archival data.
- Performed extensive literature review regarding Gamma-ray Binaries, Reverberation Mapping and AGN variability.
- Contributed my data visualization skills towards the writing of a proposal for Fermi-LAT survey.

2022 TAURUS: How do Stellar Bound Clusters Remain Together?

May - August 2022

Supervisors: Dr. Juan Pablo Farias and Dr. Stellar Offner.

- Performed N-body simulations using sophisticated [N-body6+](#) code and the Texas Advanced Computing Center [TACC].
- Analyzed physical properties of N-body simulations that took as input the most-accurate realistic magneto-hydrodynamics simulation framework [STARFORGE](#), which describes the formation of stars from GMCs.